

Activity Report
On
Door to Door Technical Assistance Reflection
And
Strategic Planning Session



Organized By:



Supported By:



NRA Gorkha



DLPIU Gorkha



Date: 4th May, 2017

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1. INTRODUCTION

1.1 Background

Nepal Earthquake of 2015 also called as The Gorkha Earthquake that struck in Nepal on April 25, 2015. About 9,000 people were killed, many thousands more were injured and more than 600,000 structures in Nepal were either damaged or destroyed. The earthquake was felt throughout central and eastern Nepal.

Effective and quality technical assistance is a major challenge for successful reconstruction process. To identify gaps, duplication, constraints in reconstruction process and to develop mutual understanding and strategy, a common platform for sharing and jointly working mechanism is needed. Technical workshop acts as an open platform for partners and government to share their evidence based learnings, best practices, challenges and a way forward for effective reconstruction process and will support for advocacy on current burning issues.

Door to door assistance reflection and strategic planning session was organized by CRS and supported by DUDBC/DLPIU, NRA and HRRP Gorkha on 4th May, 2017 to develop VDC level coordination and communication mechanism in 9 VDCs (Bungkot, Taple, Aaruarbang, Aarupokhari, Aaruchanaute, Tandrang, Manbhu, Thumi, Lapu) and field issues regarding compliance reconstruction and to develop common understanding to maintain uniformity in technical assistance and communication in each VDC.

1.2 Workshop Goal

Provide quality, effective and smooth door to door technical assistance with mutual understanding, better coordination and communication between government and partners' engineer to provide access to technical support and information for beneficiaries to get compliance and construct earthquake resilience houses.

1.3 Workshop Objectives

- To develop common/ mutual understanding between DLPIU and CRS's engineers.
- To plan for effective door to door assistance as a joint team work between government and partner's technical staff.
- To develop reporting mechanisms and track and update latest reconstruction status of VDC.
- To share the field staff experiences, challenges and learnings on the housing reconstruction

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- To collect the challenges on the technical assistance for safer reconstruction.
- To collect the ideas on defects and possible improvement from field to advocate with government.
- To establish VDC level coordination mechanism to provide technical support and to disseminate information (Technical and general) to communities.

Schedule of Technical Workshop on " Door to Door Assistance Reflection Session "		
Time		Description
8:00	8:30	Breakfast
8:30	9:00	Welcome Speech and Introduction
9:00	9:30	Speech from DUDBC Chief
9:30	10:00	Aaruchanaute Presentation (Bad Practice/Issues/Challenges), response from respective government engineers/DUDBC.
10:00	10:30	Aaruarbang Presentation (Bad Practices/Issues/Challenges) response from respective government engineers/DUDBC
10:30	10:45	Tea Break
10:45	11:15	Aarupokhari Presentation (Bad Practices/Issues/Challenges) response from respective government engineers/DUDBC
11:15	11:45	Bungkot Presentation (Bad Practices/Issues/Challenges) response from respective government engineers/DUDBC
11:45	12:15	Lapu Presentation (Bad Practices/Issues/Challenges) response from respective government engineers/DUDBC
12:15	12:45	Manbu Presentation (Bad Practices/Issues/Challenges) response from respective government engineers/DUDBC
12:45	1:45	Lunch Break
1:45	2:15	Tandrang Presentation (Bad Practices/Issues/Challenges) response from respective government engineers/DUDBC
2:15	2:45	Taple Presentation (Bad Practices/Issues/Challenges) response from respective government engineers/DUDBC
2:45	3:15	Thumi Presentation (Bad Practices/Issues/Challenges) response from respective government engineers/DUDBC
3:15	3:30	Tea Break
3:30	4:45	Coordination mechanism, discussion over action plan, reporting mechanism. (HRRP)
4:45	5:00	Conclusion and closing remarks by NRA Chief

1.4 Workshop Contents

Program was incepted by DUDBC Chief. Mr. Roshan Shrestha. He summarized the reason behind the slow pace and lack of output in current reconstruction process and briefed issues, challenges and way forward for effective and smooth reconstruction process.

- He briefed that government declared to support 3 lakhs rs as subsidy but not as full cost for reconstruction of houses along with technical support in collaboration with bilateral agencies and I/NGOs as owner driven reconstruction approach.
- Technical support is still not fully reachable to all areas due to lack of road access and difficult terrain and topography.
- Many of houses were reconstructed right after emergency phase even before Inspection guideline was published and houses constructed in first year after disaster were recorded in very different pattern. Technical support and inspection process was effectively implemented right after the release of inspection SOP (standard operating principle) and houses reconstructed after that process were constructed in very different patterns where adoption of good construction techniques were recorded.
- Availability of good quality construction materials and high transportation cost are still major challenges of reconstruction in northern VDCs of Gorkha district. For example, 10mm rod needs to be bend and rolled up to transport by mule then it will be again stretch vertically to reform as straight bar for construction. One sack of cement will cost five times more than market price and rate will increase as we go further up towards northern areas.
- RC frame houses with 9"X12" column may be safe after SAP analysis but to prevent the conflict of interest this type of house should be analysed and calculated by engineer other than inspector.
- Now, we can provide one tablet to each VDC but in every VDC there are 2 to 3 engineers and all of them must depend in single tablet which will decrease efficiency of inspection process.
- Uniformity in information and technical assistance is very important to support and influence beneficiaries to construct safe houses. Lack of coordination and uniform dissemination of information may cause trouble and risk to DLPIU engineers as they are inspectors responsible to recommend for subsidy based on inspection guideline.
- In total, we required 246 engineers but now we have only 82 engineers, 34 sub engineers and 57 Assistant sub engineers deployed in Gorkha district. We have already requested with CLPIU and NRA to fill up gaps in deployment but still we are in hard time to manage our limited human resources and in some VDC there is only one engineer.

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- Technical assistance and correction can be supported by partners' engineers but uniformity must be maintained. Coordination and communication is very necessary to develop mutual understanding between Government and partners' engineers.
- Annex 14 form for III Tranche is not enough for all VDC and even in many VDC II Tranche forms are still not enough in Gorkha District which is slow down inspection progress.
- Correction and exception manual is on progress which will sort out many non-compliance issues and support to some contradictory issues. Correction and exception manual training will roll out to government and partners' engineers but it should be published quickly.
- Government officers from district and central level commits for compliance of non-compliance houses during their field visit which will create pressure on engineers and DLPIU.
- JICA presented result of random sampling in Masel, Pandrung and Jaubari VDCs which indicates 85% of non-compliance among currently reconstructed houses. 67% non-compliance issues are due to plinth level height, practically in hilly area plinth level height are usually 5" to 6" and if we go for 100 percent accuracy 90% of houses will not be compliance with current standards.

Presentation from POs and DLPIU engineer to share bad practices, issues, challenges and lesson learned during door to door assistance in their working VDCs.

1. Aaru chanaute (Bad Practices/Issues/Challenges)

Experience sharing on Bad practices/Issues/Challenges while Door to Door Assistance. ([CLICK HERE](#))

Bad Practices:-

- Vertical posts are not placed in openings (windows and door).
- L-Shaped buildings are being constructed
- Back side of wall is constructed with stone-cement mortar and other side with brick-cement mortar
- Pillars without double strip of rods
- To place horizontal band on RCC (Junge Pillar) should be constructed but that type of pillar is not constructed.
- Hole on pillars.

Issues:-

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- Bad Practices due to poor economic condition of households.
- Initially 9/9 pillars was on the approval format of NRA but meanwhile it was changed.
- They don't think necessity of using more rods.
- Sand, Cement aggregates and water standard ratio not followed, following traditional measures

Challenges:-

- Complain received that we are only providing technical advices and not helping them on second tranche release.
- Fake information from political leaders for votes.
- Not satisfied with government speed on releasing second and third tranche release

2. Aaruarbang (Bad Practices/Issues/Challenges)

Experience sharing on Bad practices/Issues/Challenges while Door to Door Assistance. ([CLICK HERE](#))

Bad Practices:-

- Not enough pillar in building
- Roofing without roof band
- Mistakes in joints

Issues:-

- Bad practices due to economic condition of households.
- Due to lack of suitable land.
- Expensive transportation of materials

Challenges:-

- Every house wants to get tranche release but some of them is not constructed in good way.
- Fake information from political leaders for votes.
- Earthquake resistance building is not economical for people.

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3. Aarupokhari (Bad Practices/Issues/Challenges)

Experience sharing on Bad practices/Issues/Challenges while Door to Door Assistance. ([CLICK HERE](#))



Bad Practices:-

- More than 50 houses were constructed in hybrid procedure before release of inspection guideline.
- Houses are being constructing by making small footing 2/2ft and 9/9" pillars were constructed.
- DPC of thickness are 4" and horizontal bands are 3" at each 2ft of stone mud mortar wall.
- Pillars with 4 rods of 8 to 12mm dia. And single stirrups of 6mm dia @150mm were used.
- No foundation Treatment
- Improper development length and lapping of rods.
- Step wise pillar in RCC

Issues:-

- All masons involved haven't got the training regarding new construction technology.
- After earthquake people think that only pillar system houses are strong.

Challenges:-

- No involvement of youth on reconstruction activity.
- Change of trend.
- Need of people.
- Socio cultural and climatically appropriate design.

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4. Bungkot (Bad Practices/Issues/Challenges)

Experience sharing on Bad practices/Issues/Challenges while Door to Door Assistance. ([CLICK HERE](#))

Bad Practices:-

- Vertical reinforcements for openings and joints in Masonry Houses
- Poor workmanship in preparing reinforcements in RCC Houses
- Spacing of stirrups mostly 7 to 8 inches
- Low development length and lapping
- reinforcement for beam is not between column reinforcements
-

Issues:-

- Many houses have been built before publication of DUDBC catalog.
- Late publication of different designs.

Challenges:-

- Unavailability of quality construction materials.
- Poor economic condition.
- Lack of manpower

5. Lapu (Bad Practices/Issues/Challenges)

Experience sharing on Bad practices/Issues/Challenges while Door to Door Assistance. ([CLICK HERE](#))

Bad Practices:-

- Buildings without any band

Issues/ Challenges:-

- Bad Practices due to poor economic condition of households.
- Lack of Manpower and suitable land.
- Due to expensive transportation.
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6. Manbu (Bad Practices/Issues/Challenges)

Experience sharing on Bad practices/Issues/Challenges while Door to Door Assistance. ([CLICK HERE](#))

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Bad Practices:-

- Absence of plinth and roof band.
- L- shape buildings are constructed

Issues/ Challenges:-

- Bad Practices due to poor economic condition of households.
- Lack of Manpower and suitable land.
- Due to expensive transportation.

7. Tandrang (Bad Practices/Issues/Challenges)

Experience sharing on Bad practices/Issues/Challenges while Door to Door Assistance. ([CLICK HERE](#))

Bad Practices:-

- Lapping of reinforcement less than 60 dia.
- Less use of cross-stone (ie. still using small stone for filling wall)
- Misplacement of windows and lintel bands
- Use of 9"*9" sized pillar

Issues/ Challenges:-

- Political condition
- Cooperation
- Poverty
- Wrong information

8. Taple (Bad practices/Issues/Challenges)

Experience sharing on Bad practices/Issues/Challenges while Door to Door Assistance. ([CLICK HERE](#))

Bad Practices:-

- Foundation tie beam is not provided.
- Total area of building is grater then code provided.
- Overlapping of rods in joints of beam, pillars and horizontal band are not provided per code. i.e. 60D.
- The lintel of doors and windows are not in same level.

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- Vertical posts are not provided at windows and doors.
- Roofs without roof band.
- Use of timber bands without treatment.
- One side of wall is constructed with SMC and sides were constructed with BMC.

Issues/ Challenges:-

- No involvement of youth on reconstruction activity.
- Socio cultural and climatically appropriate design.
- All masons involved haven't got the training regarding new construction technology

9. Thumi (Bad practices/Issues/Challenges)

Experience sharing on Bad practices/Issues/Challenges while Door to Door Assistance. ([CLICK HERE](#))



Bad Practices:-

- Dry stone masonry up to plinth level
- length exceed more than 3-times breadth
- Big size of room, above 13.5 sq m
- Frame structural building without foundation tie beam.

Issues/ Challenges:-

- Expensive transportation of materials.
- Lack of good construction materials.
- Due to lack of mason labor

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- 10.** DUDBC vice deputy chief Mrs. Sunita Shrestha shared the major non-compliances Issues in RCC Structure and Masonry Structure. ([CLICK HERE](#))



RCC Structures (MRT):

- Single bay
- Foundation at different levels
- Pillars of 12x12 size but with 8 Nos of 12 dia bars
- Span of room exceeding 4.5 meter
- Floor area of room exceeding 13.5 m²
- Absence foundation beam
- Use of 9x9 or 9x12 pillars
- RCC pillars in 9"x9" but roof is with CGI sheet.
- Shape of building-projections exceeding 1/6th of minimum width of building

Masonry structures:

- Stone masonry in foundation and brick masonry in super structure.
- Span of room exceeding 4.5 meter
- Floor area of room exceeding 13.5 m²
- Vertical reinforcement bars only in corners and joints but not in the sides of openings
- Masonry houses, due to passage in between, opening requirements is not fulfilled
- Up to plinth level stone masonry but above plinth band use of steel structure (iron post and truss) or timber structure (timber posts with planking and timber truss)

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- Ground floor stone or brick masonry with timber bands but having attic of timber frame with CGI sheets

11. Joint coordination and reporting mechanism-HRRP, Gorkha

- HRRP district coordinator Mr. Minar Thapa Magar presented the Correction/Exception Manual for Masonry Structure published with purpose to provide corrective measures to non-compliance houses to be safe and get compliance. This manual will also provide solution to major issues and exceptions observed during reconstruction of houses at field level. He updated that Training of trainers ToT will be organized at central and district level to train government and partners' engineer. ([Click Here](#))
- HRRP also presented RCC non-compliance check and analyzed design of 11 buildings which violates single mandatory rule ([Click Here](#)) and DUDBC design catalogue volume-II for alternative materials. ([Click Here](#))
- He facilitated joint coordination and reporting mechanism session which was incepted with brief lecture on need of dissemination of information (technical and general), technical support during construction and door to door technical assistance. He also urged participants to focus more on developing mutual respect and trust between government and partners' engineer and beneficiaries.
- Joint coordination mechanism was developed for each VDC by participants through group work. Some key points decided are as follows:
 1. Frequent communication through telephone and other mediums.
 2. Information sharing and technical support through partnership with a good team work.
 3. Government engineers will coordinate and support partner's engineers in gaining community trust by accompany them in inspection and VDC level sharing program.
 4. Regular VDC level monthly coordination meeting in which simple monthly report will be produced by partners' engineers which will reflect ward level reconstruction status and data. These deliverables will be shared with NRA, DUDBC and HRRP on regular basis.

NRA Gorkha representative gave his vote of thanks for organizing door to door technical assistance reflection and strategic planning session and creating platform to develop mutual understanding. He summarized the reason behind the delay of reconstruction process and its timeline with major learnings and achievements and requested everyone's support to make it happen.

- III Tranche forms are ready for printing but due to lack of resources CLPIU are unable to print it in time.

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- Government engineers must introduce door to door technical assistance engineer to communities to strengthen mutual respect and trust which could result in effective and smooth technical assistance.
- Communication is key for effective coordination and influence in communities and NRA requested participants to maintain professional standards of behavior and communicate with beneficiaries as much as possible in a polite and vigilant manner.
- He also directed that partners should adopt relevant and effective selection criteria so that authentic and real mason will be trained who are willing to contribute in reconstruction of houses.
- NRA requested participants to communicate with NRA via email (nragorkha@gmail.com) or directly by call.

Annex- 2: Photographs



Annex 1: Registration of Participants

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